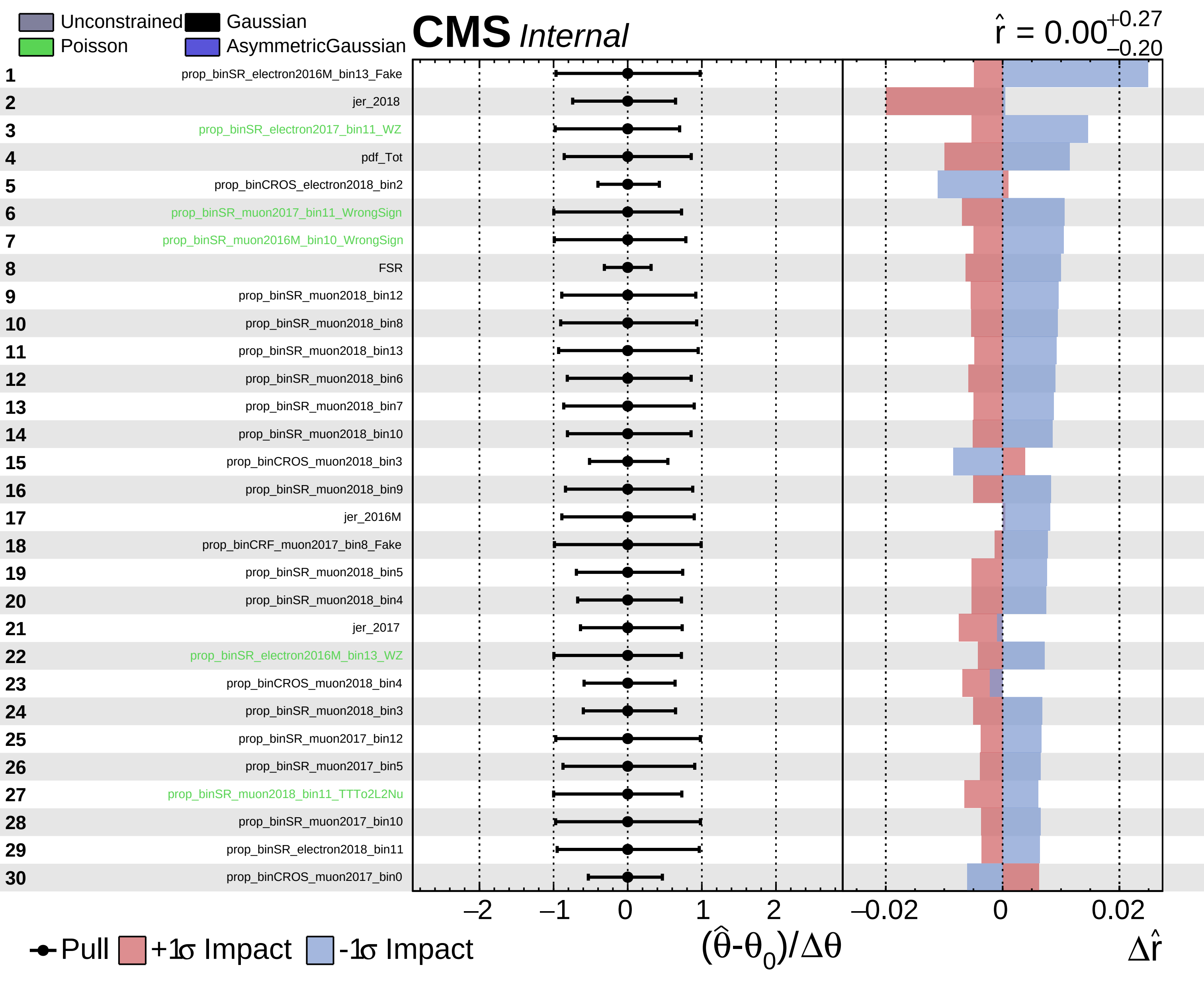
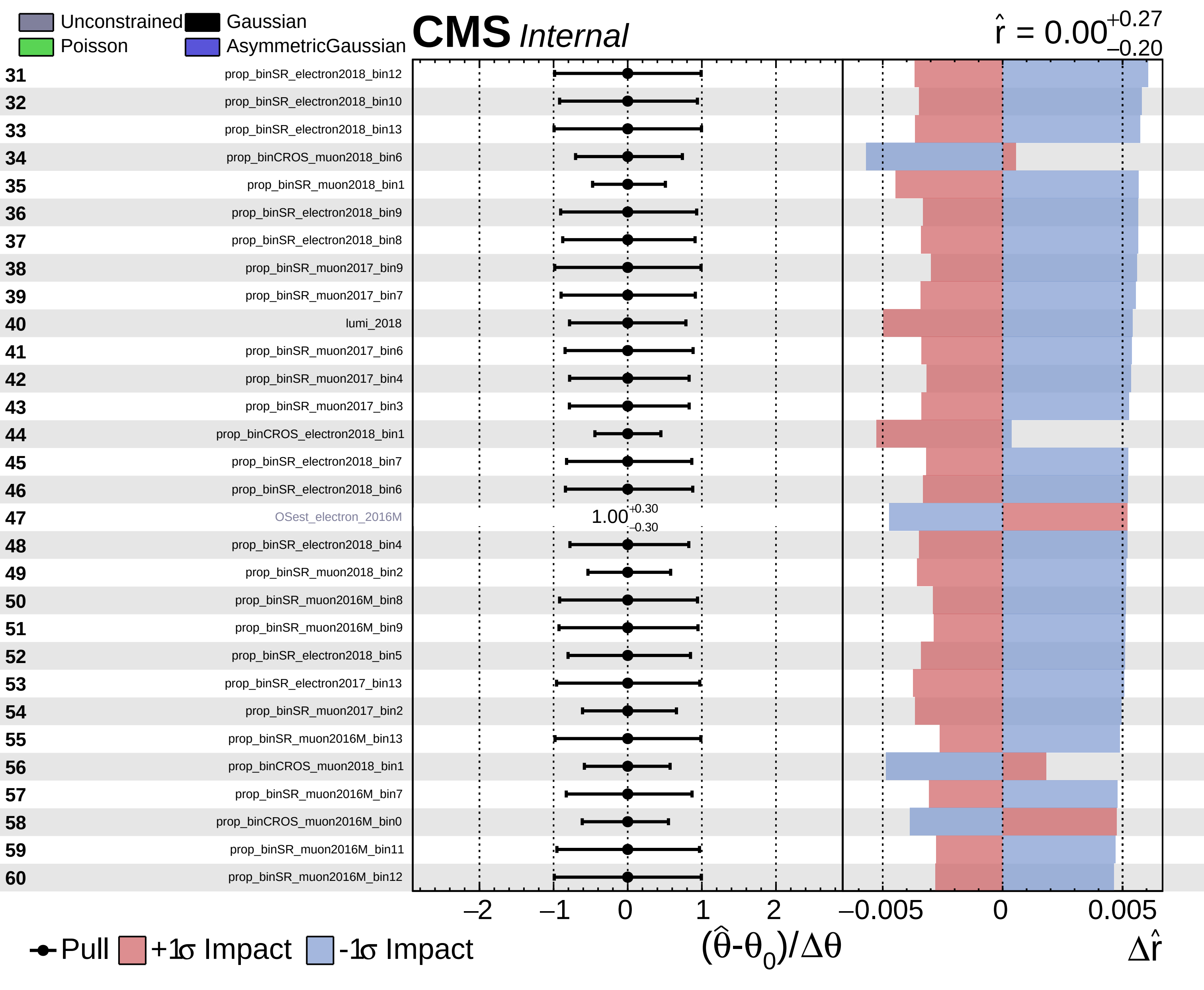
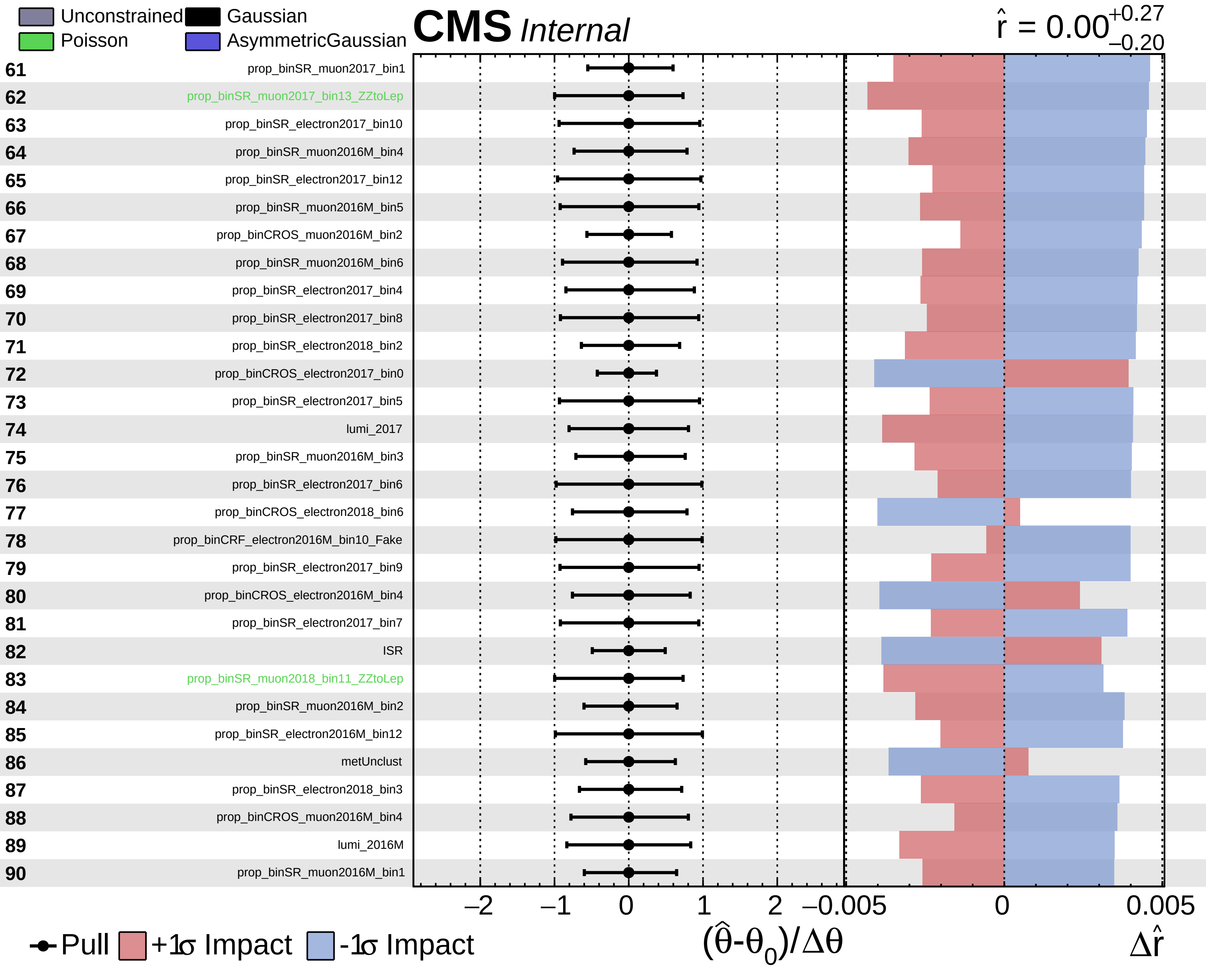
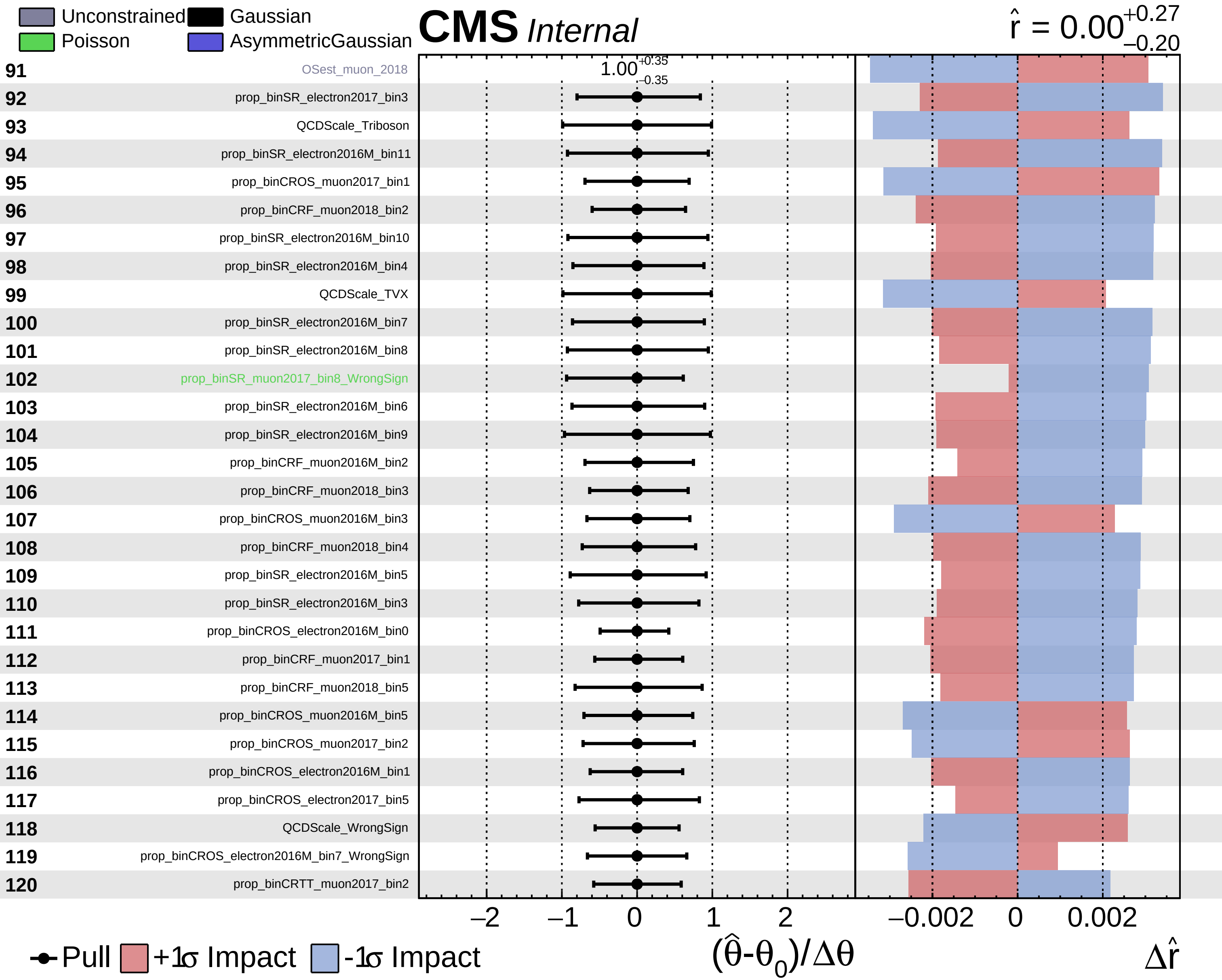

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
1 prop_binSR_electron2016M_bin13_Fake
2 jer_2018
3 prop_binSR_electron2017_bin11_WZ
4 pdf_Tot
5 prop_binCROS_electron2018_bin2
6 prop_binSR_muon2017_bin11_WrongSign
7 prop_binSR_muon2016M_bin10_WrongSign
8 FSR
9 prop_binSR_muon2018_bin12
10 prop_binSR_muon2018_bin8
11 prop_binSR_muon2018_bin13
12 prop_binSR_muon2018_bin6
13 prop_binSR_muon2018_bin7
14 prop_binSR_muon2018_bin10
15 prop_binCROS_muon2018_bin3
16 prop_binSR_muon2018_bin9
17 jer_2016M
18 prop_binCRF_muon2017_bin8_Fake
19 prop_binSR_muon2018_bin5
20 prop_binSR_muon2018_bin4
21 jer_2017
22 prop_binSR_electron2016M_bin13_WZ
23 prop_binCROS_muon2018_bin4
24 prop_binSR_muon2018_bin3
25 prop_binSR_muon2017_bin12
26 prop_binSR_muon2017_bin5
27 prop_binSR_muon2018_bin11_TTTo2L2Nu
28 prop_binSR_muon2017_bin10
29 prop_binSR_electron2018_bin11
30 prop_binCROS_muon2017_bin0
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



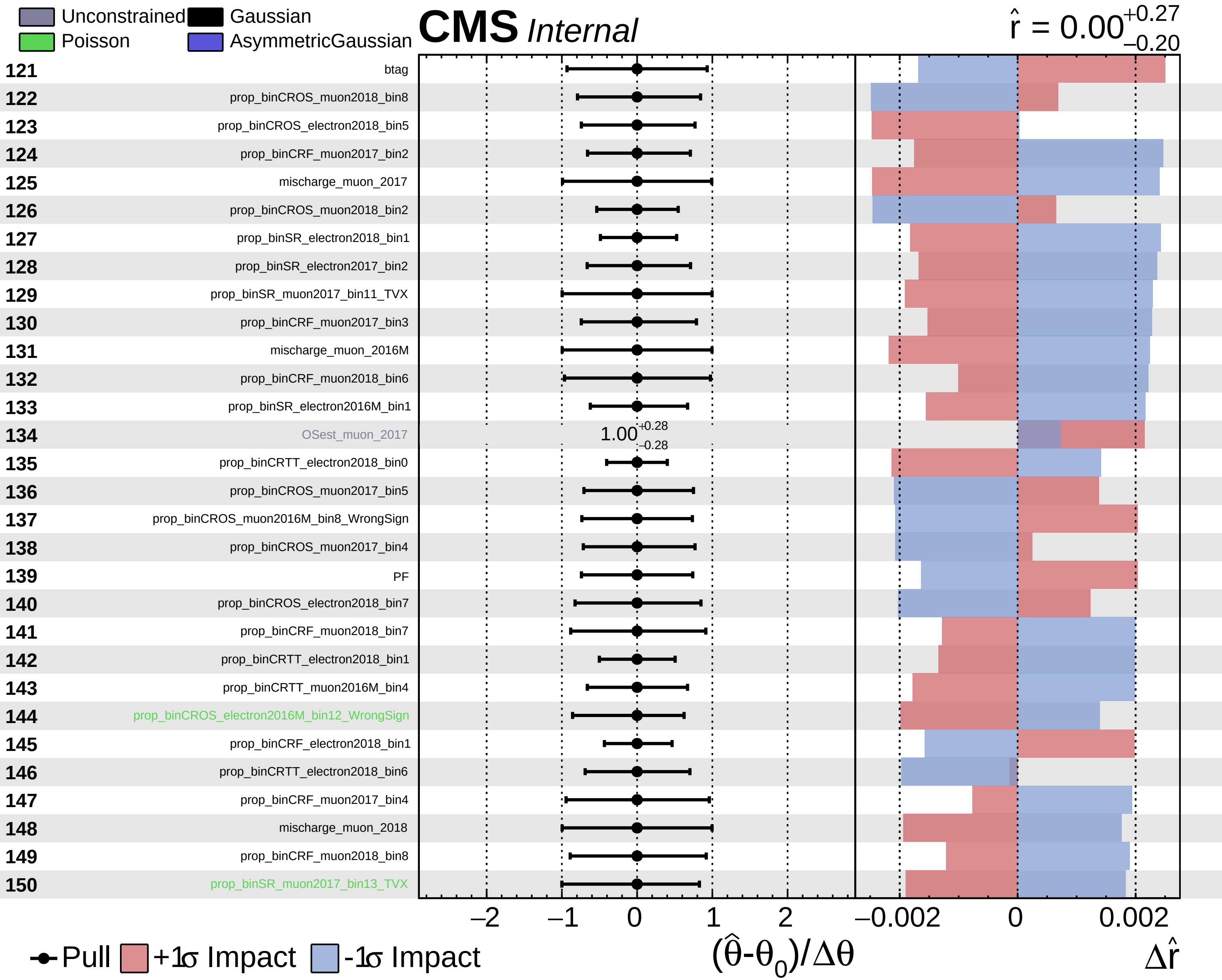
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
31 prop_binSR_electron2018_bin12
32 prop_binSR_electron2018_bin10
33 prop_binSR_electron2018_bin13
34 prop_binCROS_muon2018_bin6
35 prop_binSR_muon2018_bin1
36 prop_binSR_electron2018_bin9
37 prop_binSR_electron2018_bin8
38 prop_binSR_muon2017_bin9
39 prop_binSR_muon2017_bin7
40 lumi_2018
41 prop_binSR_muon2017_bin6
42 prop_binSR_muon2017_bin4
43 prop_binSR_muon2017_bin3
44 prop_binCROS_electron2018_bin1
45 prop_binSR_electron2018_bin7
46 prop_binSR_electron2018_bin6
47 OSest_electron_2016M
$1.00^{+0.30}_{-0.30}$
48 prop_binSR_electron2018_bin4
49 prop_binSR_muon2018_bin2
50 prop_binSR_muon2016M_bin8
51 prop_binSR_muon2016M_bin9
52 prop_binSR_electron2018_bin5
53 prop_binSR_electron2017_bin13
54 prop_binSR_muon2017_bin2
55 prop_binSR_muon2016M_bin13
56 prop_binCROS_muon2018_bin1
57 prop_binSR_muon2016M_bin7
58 prop_binCROS_muon2016M_bin0
59 prop_binSR_muon2016M_bin11
60 prop_binSR_muon2016M_bin12
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



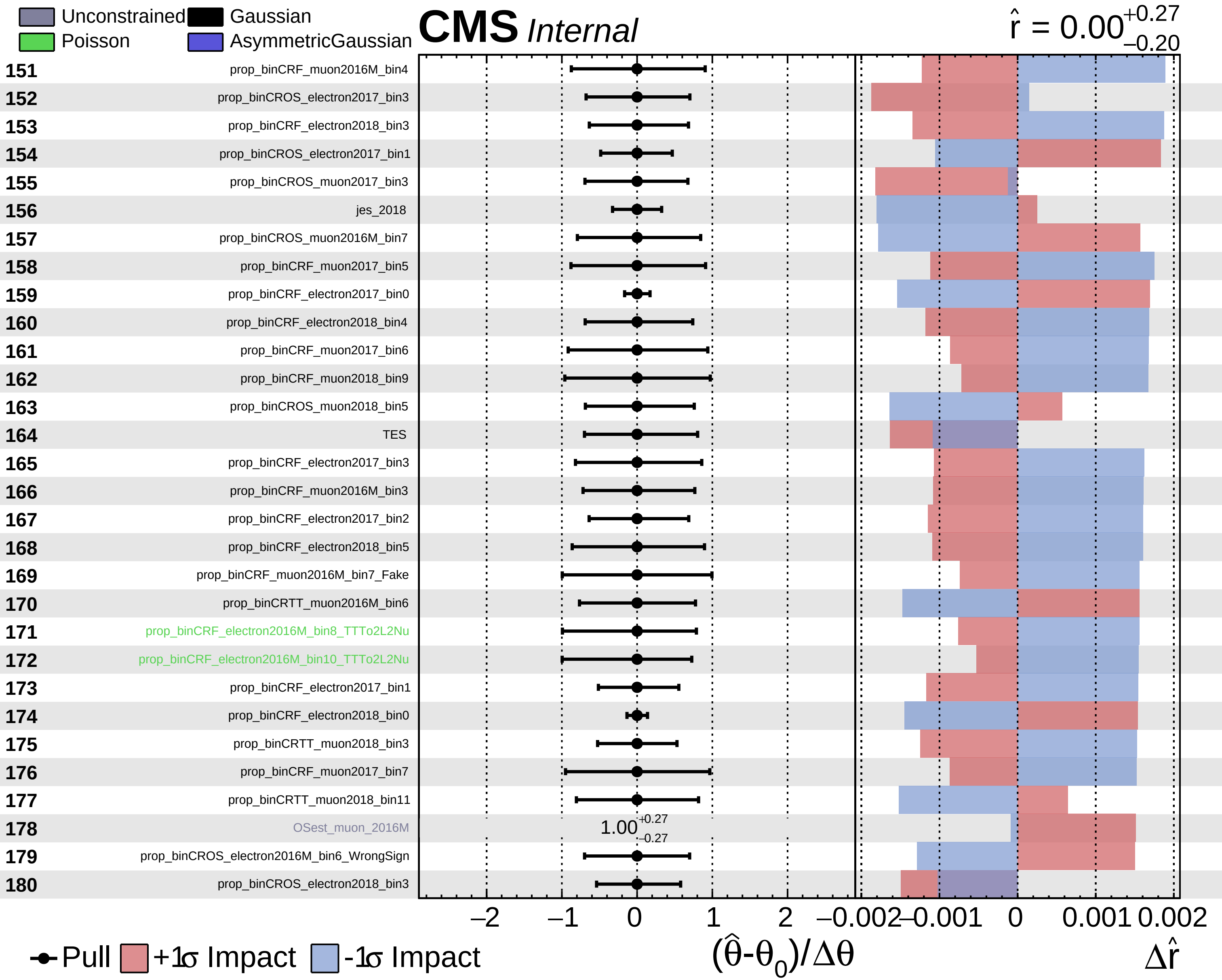
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
61 prop_binSR_muon2017_bin1
62 prop_binSR_muon2017_bin13_ZZtoLep
63 prop_binSR_electron2017_bin10
64 prop_binSR_muon2016M_bin4
65 prop_binSR_electron2017_bin12
66 prop_binSR_muon2016M_bin5
67 prop_binCROS_muon2016M_bin2
68 prop_binSR_muon2016M_bin6
69 prop_binSR_electron2017_bin4
70 prop_binSR_electron2017_bin8
71 prop_binSR_electron2018_bin2
72 prop_binCROS_electron2017_bin0
73 prop_binSR_electron2017_bin5
74 lumi_2017
75 prop_binSR_muon2016M_bin3
76 prop_binSR_electron2017_bin6
77 prop_binCROS_electron2018_bin6
78 prop_binCRF_electron2016M_bin10_Fake
79 prop_binSR_electron2017_bin9
80 prop_binCROS_electron2016M_bin4
81 prop_binSR_electron2017_bin7
82 ISR
83 prop_binSR_muon2018_bin11_ZZtoLep
84 prop_binSR_muon2016M_bin2
85 prop_binSR_electron2016M_bin12
86 metUnclust
87 prop_binSR_electron2018_bin3
88 prop_binCROS_muon2016M_bin4
89 lumi_2016M
90 prop_binSR_muon2016M_bin1
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



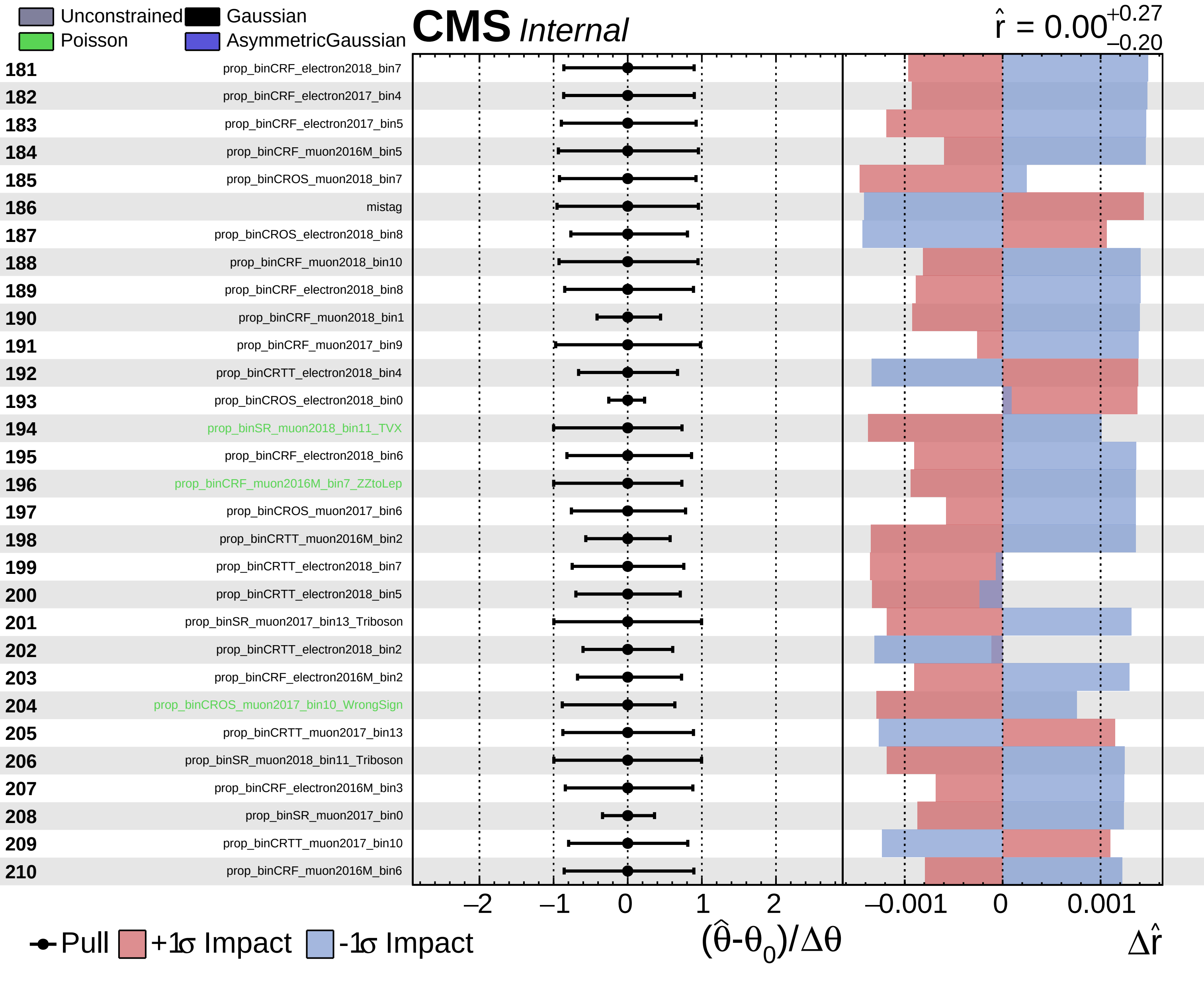
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
91 OSest_muon_2018
$1.00^{+0.35}_{-0.35}$
92 prop_binSR_electron2017_bin3
93 QCDScale_Triboson
94 prop_binSR_electron2016M_bin11
95 prop_binCROS_muon2017_bin1
96 prop_binCRF_muon2018_bin2
97 prop_binSR_electron2016M_bin10
98 prop_binSR_electron2016M_bin4
99 QCDScale_TVX
100 prop_binSR_electron2016M_bin7
101 prop_binSR_electron2016M_bin8
102 prop_binSR_muon2017_bin8_WrongSign
103 prop_binSR_electron2016M_bin6
104 prop_binSR_electron2016M_bin9
105 prop_binCRF_muon2016M_bin2
106 prop_binCRF_muon2018_bin3
107 prop_binCROS_muon2016M_bin3
108 prop_binCRF_muon2018_bin4
109 prop_binSR_electron2016M_bin5
110 prop_binSR_electron2016M_bin3
111 prop_binCROS_electron2016M_bin0
112 prop_binCRF_muon2017_bin1
113 prop_binCRF_muon2018_bin5
114 prop_binCROS_muon2016M_bin5
115 prop_binCROS_muon2017_bin2
116 prop_binCROS_electron2016M_bin1
117 prop_binCROS_electron2017_bin5
118 QCDScale_WrongSign
119 prop_binCROS_electron2016M_bin7_WrongSign
120 prop_binCRTT_muon2017_bin2
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
121 btag
122 prop_binCROS_muon2018_bin8
123 prop_binCROS_electron2018_bin5
124 prop_binCRF_muon2017_bin2
125 mischarge_muon_2017
126 prop_binCROS_muon2018_bin2
127 prop_binSR_electron2018_bin1
128 prop_binSR_electron2017_bin2
129 prop_binSR_muon2017_bin11_TVX
130 prop_binCRF_muon2017_bin3
131 mischarge_muon_2016M
132 prop_binCRF_muon2018_bin6
133 prop_binSR_electron2016M_bin1
134 OSest_muon_2017
$1.00^{+0.28}_{-0.28}$
135 prop_binCRTT_electron2018_bin0
136 prop_binCROS_muon2017_bin5
137 prop_binCROS_muon2016M_bin8_WrongSign
138 prop_binCROS_muon2017_bin4
139 PF
140 prop_binCROS_electron2018_bin7
141 prop_binCRF_muon2018_bin7
142 prop_binCRTT_electron2018_bin1
143 prop_binCRTT_muon2016M_bin4
144 prop_binCROS_electron2016M_bin12_WrongSign
145 prop_binCRF_electron2018_bin1
146 prop_binCRTT_electron2018_bin6
147 prop_binCRF_muon2017_bin4
148 mischarge_muon_2018
149 prop_binCRF_muon2018_bin8
150 prop_binSR_muon2017_bin13_TVX
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
151 prop_binCRF_muon2016M_bin4
152 prop_binCROS_electron2017_bin3
153 prop_binCRF_electron2018_bin3
154 prop_binCROS_electron2017_bin1
155 prop_binCROS_muon2017_bin3
156 jes_2018
157 prop_binCROS_muon2016M_bin7
158 prop_binCRF_muon2017_bin5
159 prop_binCRF_electron2017_bin0
160 prop_binCRF_electron2018_bin4
161 prop_binCRF_muon2017_bin6
162 prop_binCRF_muon2018_bin9
163 prop_binCROS_muon2018_bin5
164 TES
165 prop_binCRF_electron2017_bin3
166 prop_binCRF_muon2016M_bin3
167 prop_binCRF_electron2017_bin2
168 prop_binCRF_electron2018_bin5
169 prop_binCRF_muon2016M_bin7_Fake
170 prop_binCRTT_muon2016M_bin6
171 prop_binCRF_electron2016M_bin8_TTTo2L2Nu
172 prop_binCRF_electron2016M_bin10_TTTo2L2Nu
173 prop_binCRF_electron2017_bin1
174 prop_binCRF_electron2018_bin0
175 prop_binCRTT_muon2018_bin3
176 prop_binCRF_muon2017_bin7
177 prop_binCRTT_muon2018_bin11
178 OSest_muon_2016M
$1.00^{+0.27}_{-0.27}$
179 prop_binCROS_electron2016M_bin6_WrongSign
180 prop_binCROS_electron2018_bin3
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
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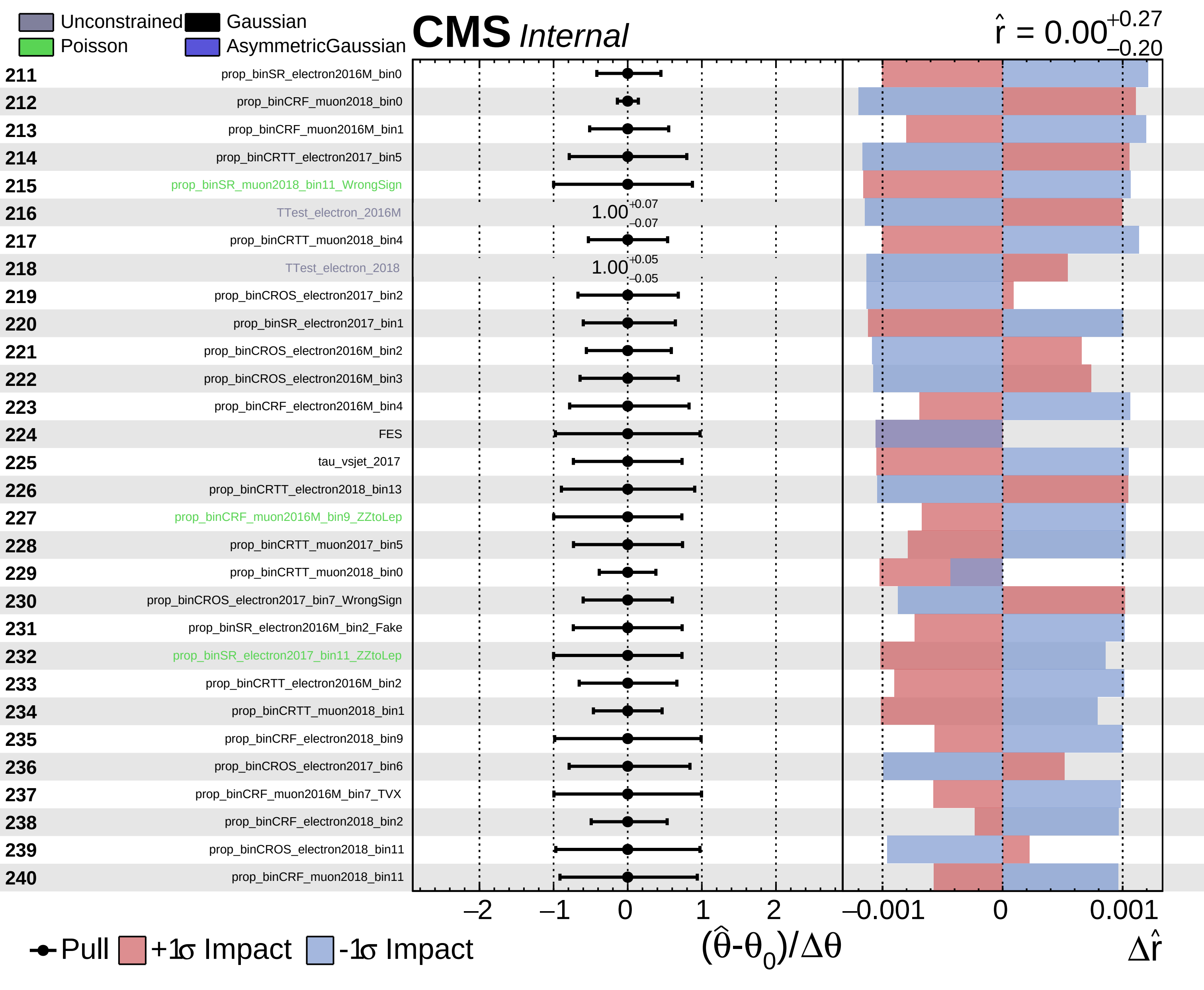
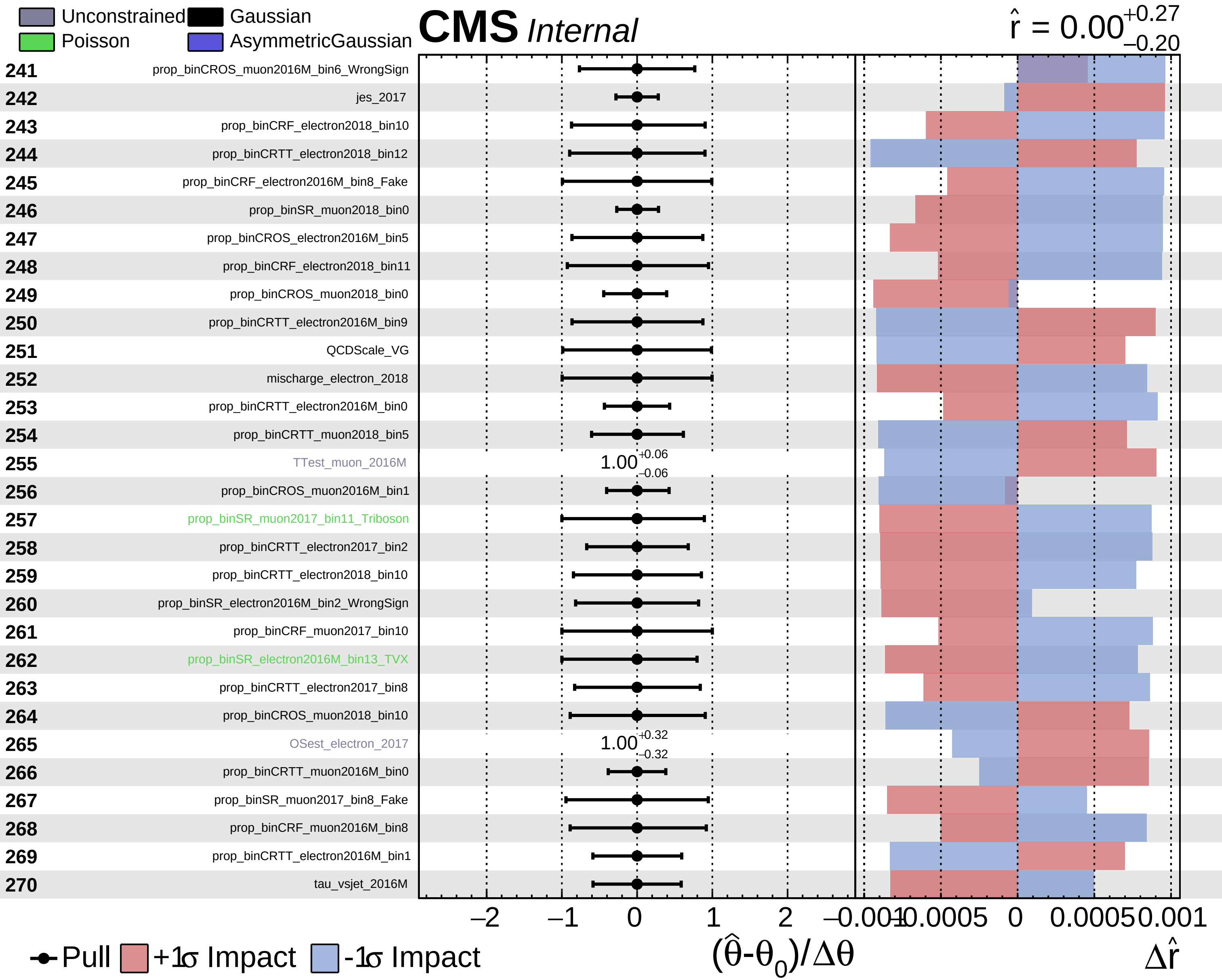
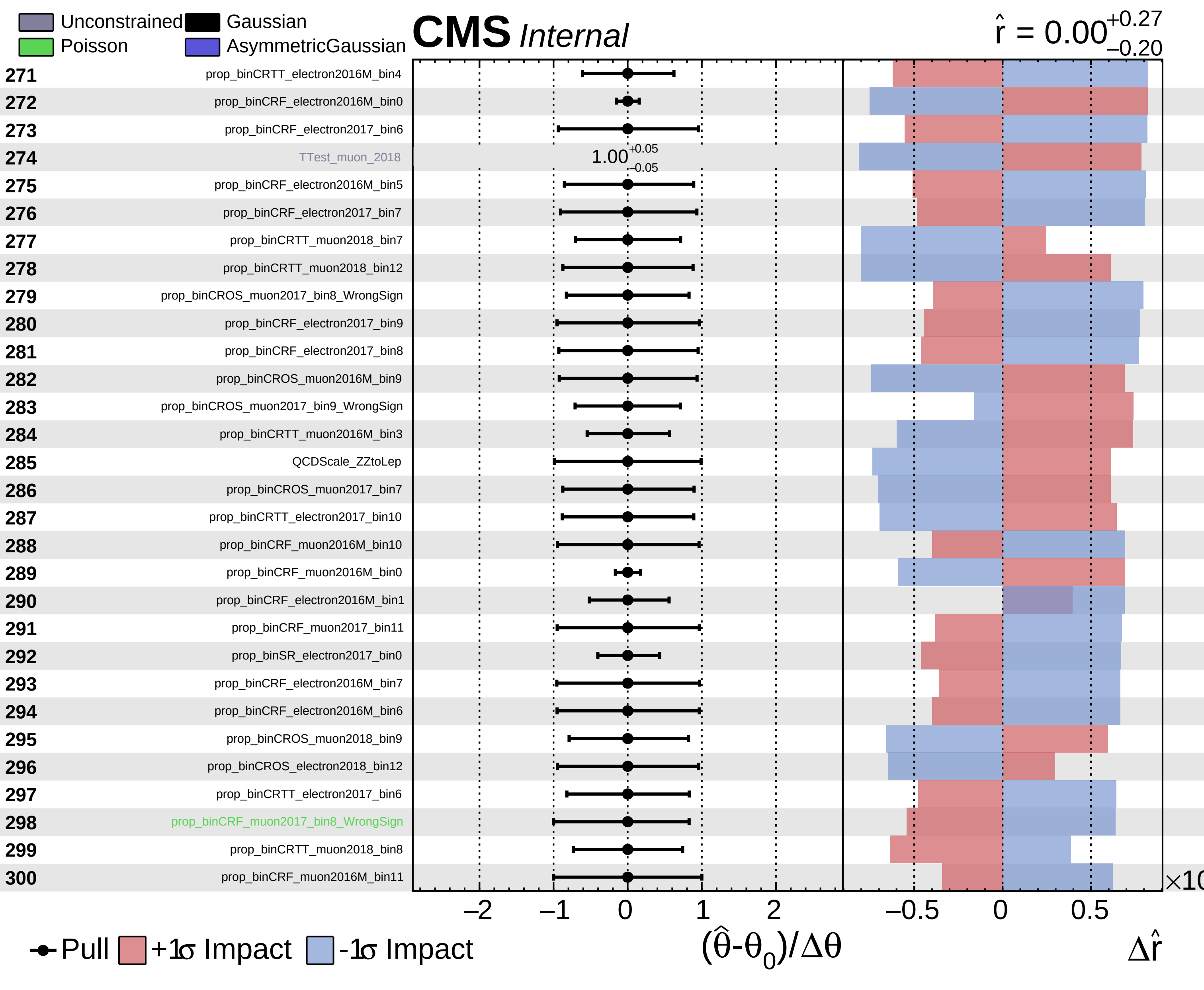
182 prop_binCRF_electron2017_bin4
183 prop_binCRF_electron2017_bin5
184 prop_binCRF_muon2016M_bin5
185 prop_binCROS_muon2018_bin7
186 mistag
187 prop_binCROS_electron2018_bin8
188 prop_binCRF_muon2018_bin10
189 prop_binCRF_electron2018_bin8
190 prop_binCRF_muon2018_bin1
191 prop_binCRF_muon2017_bin9
192 prop_binCRTT_electron2018_bin4
193 prop_binCROS_electron2018_bin0
194 prop_binSR_muon2018_bin11_TVX
195 prop_binCRF_electron2018_bin6
196 prop_binCRF_muon2016M_bin7_ZZtoLep
197 prop_binCROS_muon2017_bin6
198 prop_binCRTT_muon2016M_bin2
199 prop_binCRTT_electron2018_bin7
200 prop_binCRTT_electron2018_bin5
201 prop_binSR_muon2017_bin13_Triboson
202 prop_binCRTT_electron2018_bin2
203 prop_binCRF_electron2016M_bin2
204 prop_binCROS_muon2017_bin10_WrongSign
205 prop_binCRTT_muon2017_bin13
206 prop_binSR_muon2018_bin11_Triboson
207 prop_binCRF_electron2016M_bin3
208 prop_binSR_muon2017_bin0
209 prop_binCRTT_muon2017_bin10
210 prop_binCRF_muon2016M_bin6
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



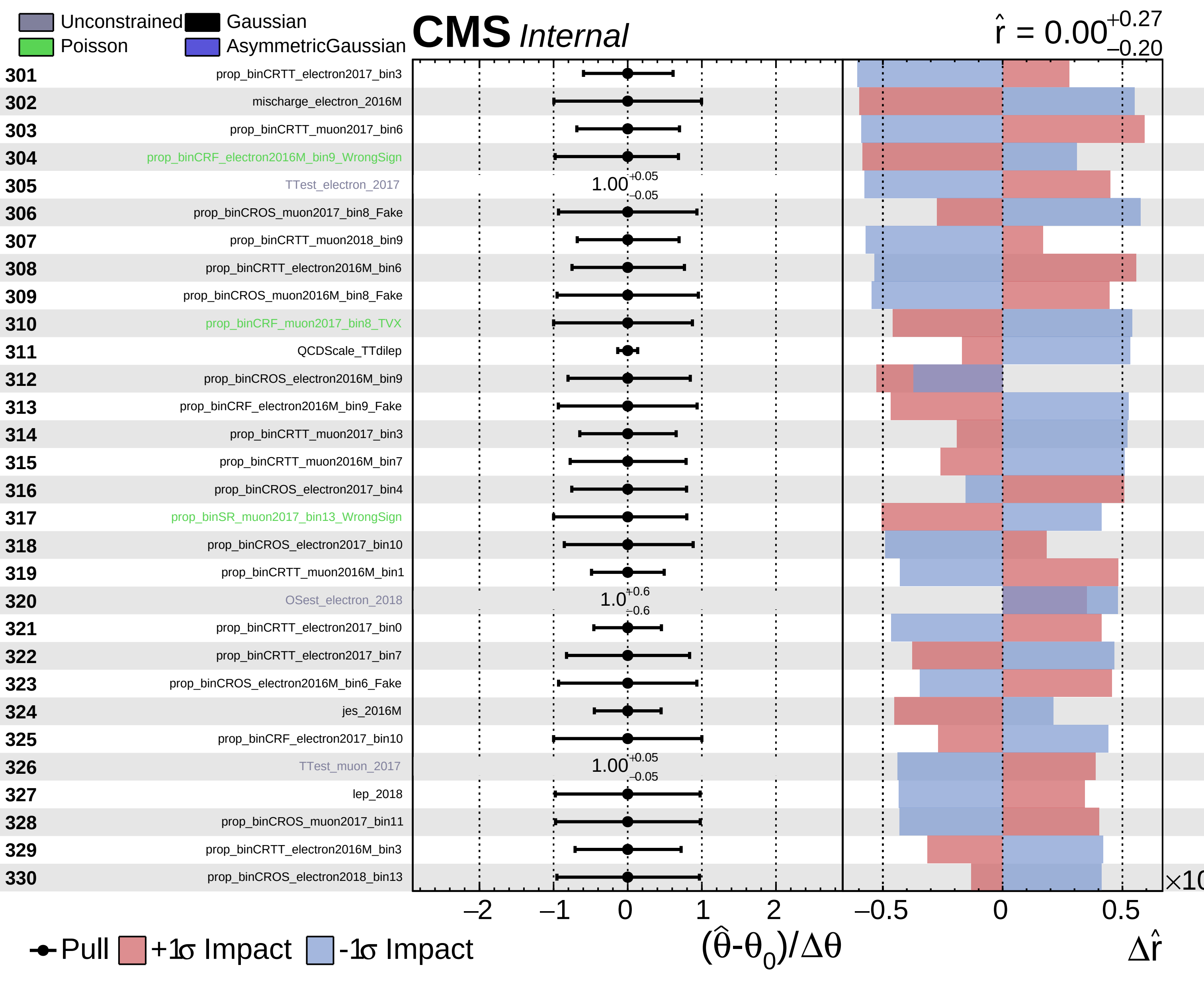
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
211 prop_binSR_electron2016M_bin0
212 prop_binCRF_muon2018_bin0
213 prop_binCRF_muon2016M_bin1
214 prop_binCRTT_electron2017_bin5
215 prop_binSR_muon2018_bin11_WrongSign
216 TTest_electron_2016M
$1.00^{+0.07}_{-0.07}$
217 prop_binCRTT_muon2018_bin4
218 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
219 prop_binCROS_electron2017_bin2
220 prop_binSR_electron2017_bin1
221 prop_binCROS_electron2016M_bin2
222 prop_binCROS_electron2016M_bin3
223 prop_binCRF_electron2016M_bin4
224 FES
225 tau_vsjet_2017
226 prop_binCRTT_electron2018_bin13
227 prop_binCRF_muon2016M_bin9_ZZtoLep
228 prop_binCRTT_muon2017_bin5
229 prop_binCRTT_muon2018_bin0
230 prop_binCROS_electron2017_bin7_WrongSign
231 prop_binSR_electron2016M_bin2_Fake
232 prop_binSR_electron2017_bin11_ZZtoLep
233 prop_binCRTT_electron2016M_bin2
234 prop_binCRTT_muon2018_bin1
235 prop_binCRF_electron2018_bin9
236 prop_binCROS_electron2017_bin6
237 prop_binCRF_muon2016M_bin7_TVX
238 prop_binCRF_electron2018_bin2
239 prop_binCROS_electron2018_bin11
240 prop_binCRF_muon2018_bin11
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



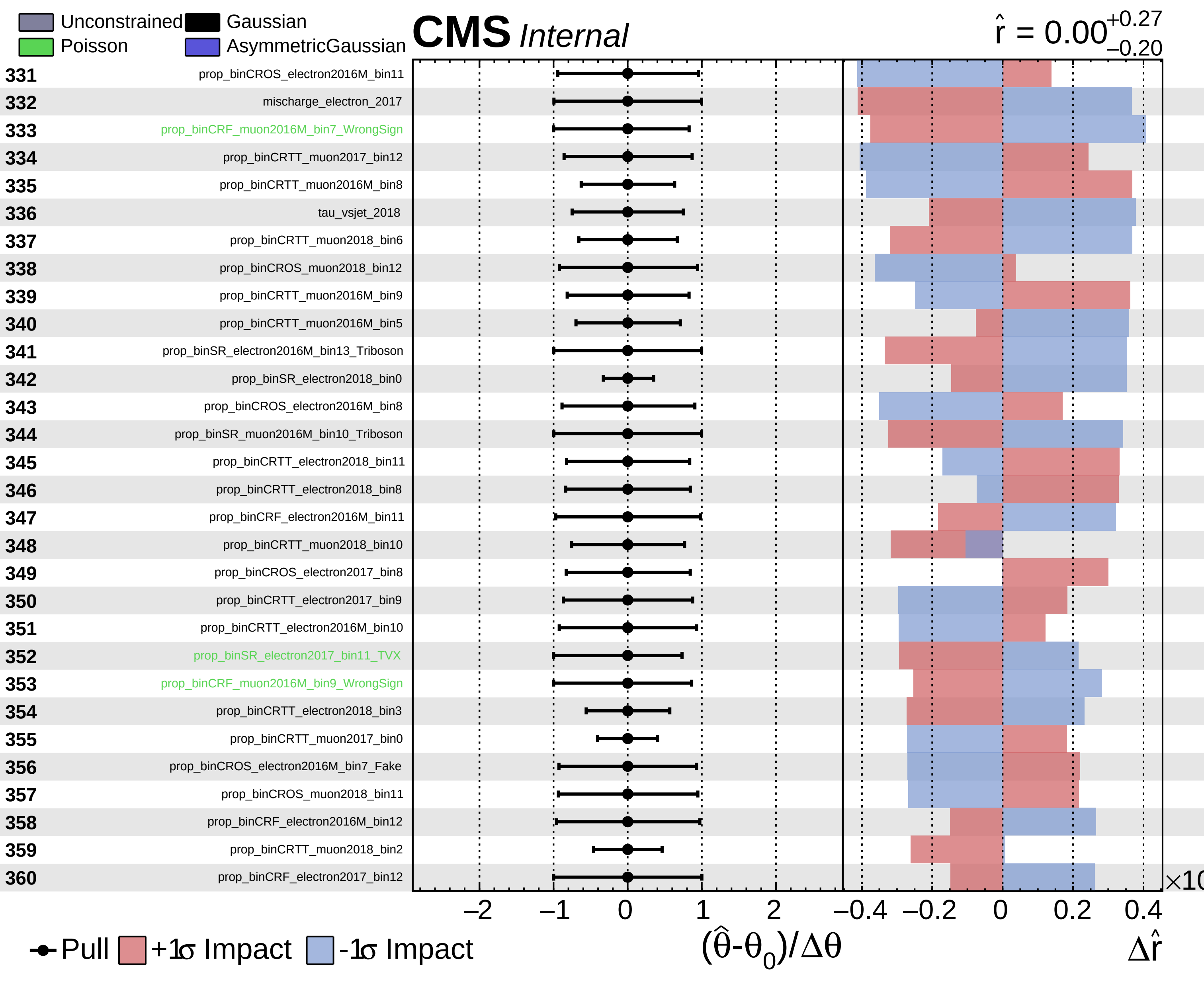
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
241 prop_binCROS_muon2016M_bin6_WrongSign
242 jes_2017
243 prop_binCRF_electron2018_bin10
244 prop_binCRTT_electron2018_bin12
245 prop_binCRF_electron2016M_bin8_Fake
246 prop_binSR_muon2018_bin0
247 prop_binCROS_electron2016M_bin5
248 prop_binCRF_electron2018_bin11
249 prop_binCROS_muon2018_bin0
250 prop_binCRTT_electron2016M_bin9
251 QCDScale_VG
252 mischarge_electron_2018
253 prop_binCRTT_electron2016M_bin0
254 prop_binCRTT_muon2018_bin5
255 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
256 prop_binCROS_muon2016M_bin1
257 prop_binSR_muon2017_bin11_Triboson
258 prop_binCRTT_electron2017_bin2
259 prop_binCRTT_electron2018_bin10
260 prop_binSR_electron2016M_bin2_WrongSign
261 prop_binCRF_muon2017_bin10
262 prop_binSR_electron2016M_bin13_TVX
263 prop_binCRTT_electron2017_bin8
264 prop_binCROS_muon2018_bin10
265 OSest_electron_2017
$1.00^{+0.32}_{-0.32}$
266 prop_binCRTT_muon2016M_bin0
267 prop_binSR_muon2017_bin8_Fake
268 prop_binCRF_muon2016M_bin8
269 prop_binCRTT_electron2016M_bin1
270 tau_vsjet_2016M
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

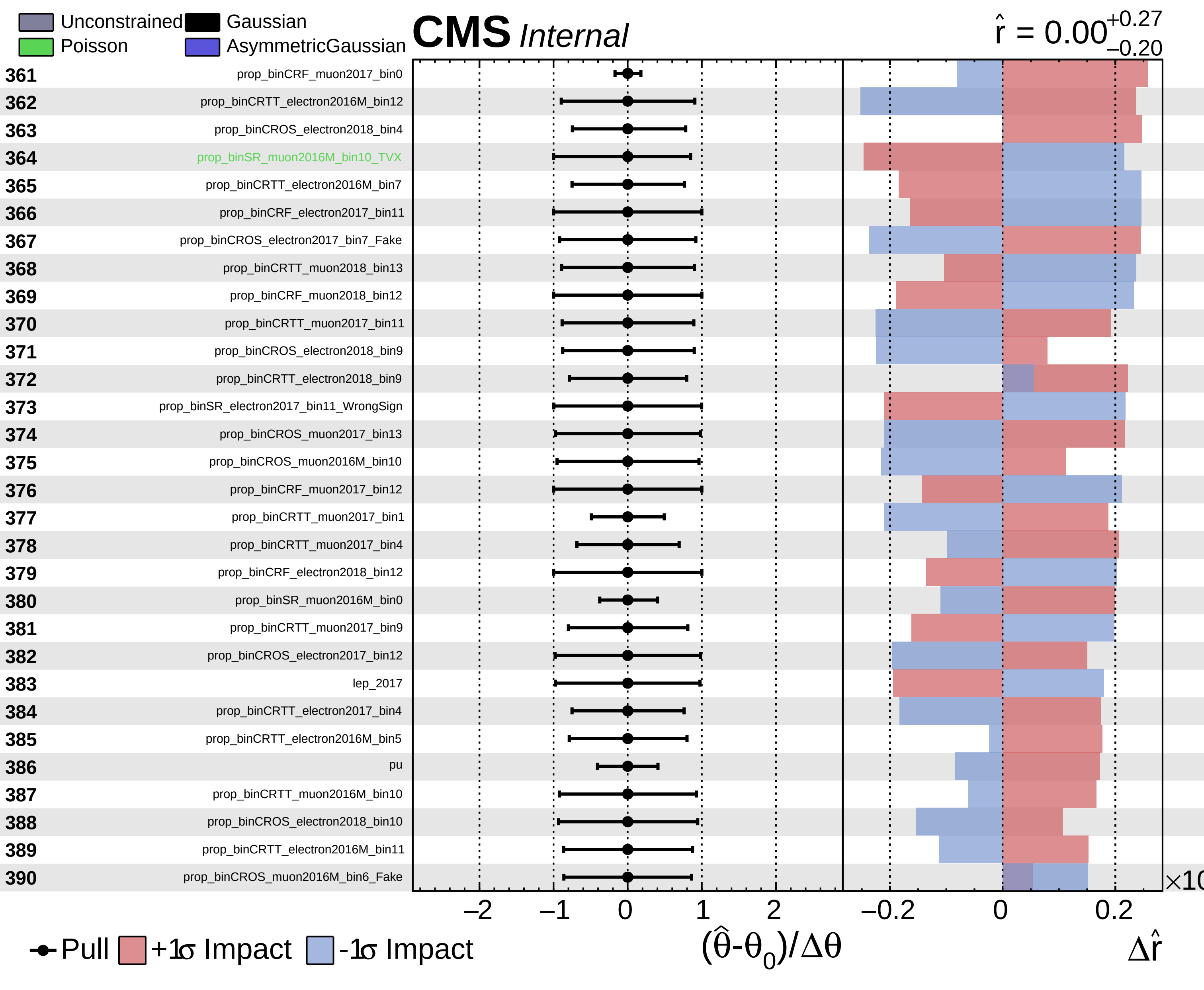


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
271 prop_binCRTT_electron2016M_bin4
272 prop_binCRF_electron2016M_bin0
273 prop_binCRF_electron2017_bin6
274 TTest_muon_2018 $1.00^{+0.05}_{-0.05}$
275 prop_binCRF_electron2016M_bin5
276 prop_binCRF_electron2017_bin7
277 prop_binCRTT_muon2018_bin7
278 prop_binCRTT_muon2018_bin12
279 prop_binCROS_muon2017_bin8_WrongSign
280 prop_binCRF_electron2017_bin9
281 prop_binCRF_electron2017_bin8
282 prop_binCROS_muon2016M_bin9
283 prop_binCROS_muon2017_bin9_WrongSign
284 prop_binCRTT_muon2016M_bin3
285 QCDScale_ZZtoLep
286 prop_binCROS_muon2017_bin7
287 prop_binCRTT_electron2017_bin10
288 prop_binCRF_muon2016M_bin10
289 prop_binCRF_muon2016M_bin0
290 prop_binCRF_electron2016M_bin1
291 prop_binCRF_muon2017_bin11
292 prop_binSR_electron2017_bin0
293 prop_binCRF_electron2016M_bin7
294 prop_binCRF_electron2016M_bin6
295 prop_binCROS_muon2018_bin9
296 prop_binCROS_electron2018_bin12
297 prop_binCRTT_electron2017_bin6
298 prop_binCRF_muon2017_bin8_WrongSign
299 prop_binCRTT_muon2018_bin8
300 prop_binCRF_muon2016M_bin11
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

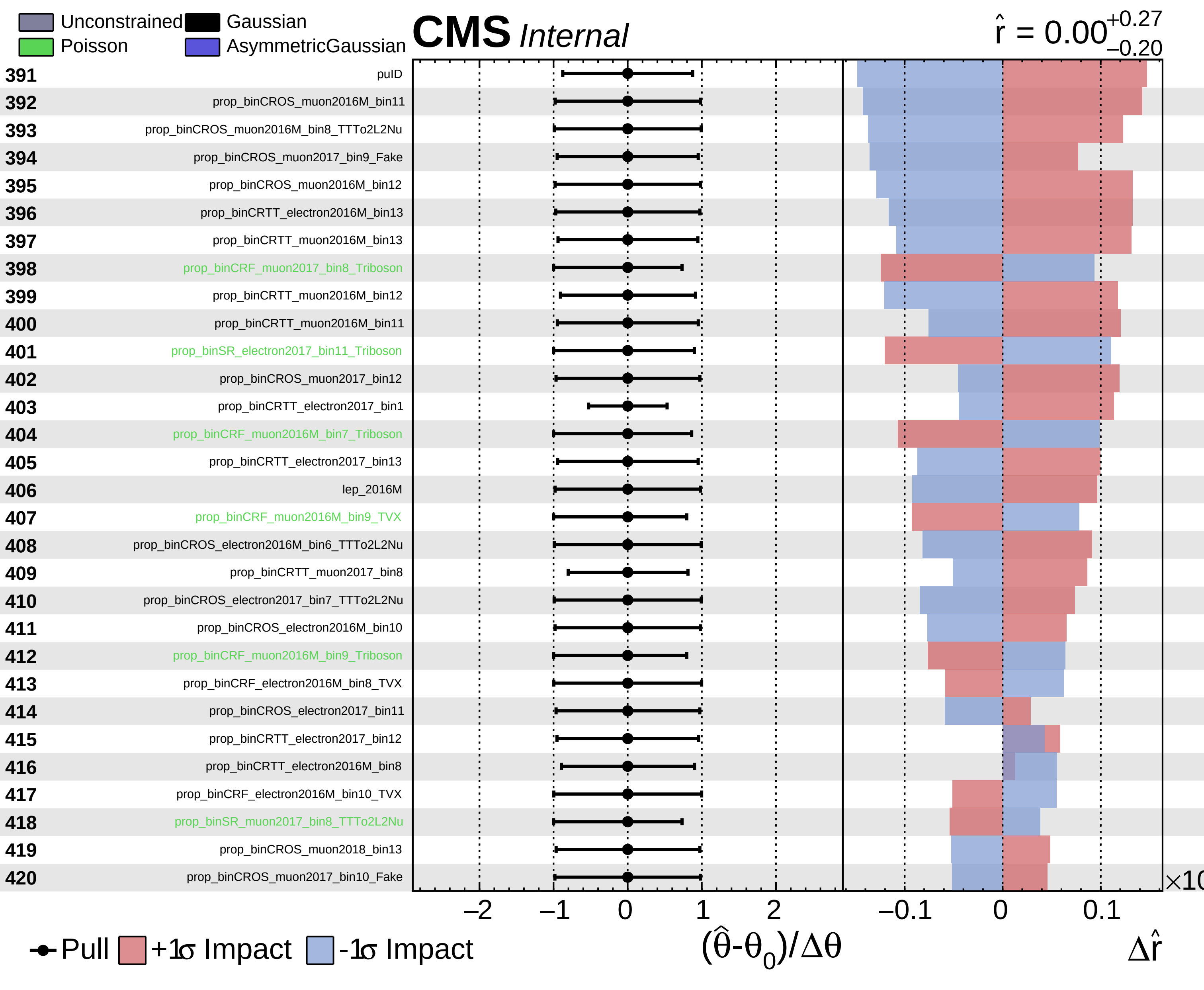


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
301 prop_binCRTT_electron2017_bin3
302 mischarge_electron_2016M
303 prop_binCRTT_muon2017_bin6
304 prop_binCRF_electron2016M_bin9_WrongSign
305 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
306 prop_binCROS_muon2017_bin8_Fake
307 prop_binCRTT_muon2018_bin9
308 prop_binCRTT_electron2016M_bin6
309 prop_binCROS_muon2016M_bin8_Fake
310 prop_binCRF_muon2017_bin8_TVX
311 QCDScale_TTdilep
312 prop_binCROS_electron2016M_bin9
313 prop_binCRF_electron2016M_bin9_Fake
314 prop_binCRTT_muon2017_bin3
315 prop_binCRTT_muon2016M_bin7
316 prop_binCROS_electron2017_bin4
317 prop_binSR_muon2017_bin13_WrongSign
318 prop_binCROS_electron2017_bin10
319 prop_binCRTT_muon2016M_bin1
320 OSest_electron_2018
$1.0^{+0.6}_{-0.6}$
321 prop_binCRTT_electron2017_bin0
322 prop_binCRTT_electron2017_bin7
323 prop_binCROS_electron2016M_bin6_Fake
324 jes_2016M
325 prop_binCRF_electron2017_bin10
326 TTest_muon_2017
$1.00^{+0.05}_{-0.05}$
327 lep_2018
328 prop_binCROS_muon2017_bin11
329 prop_binCRTT_electron2016M_bin3
330 prop_binCROS_electron2018_bin13
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

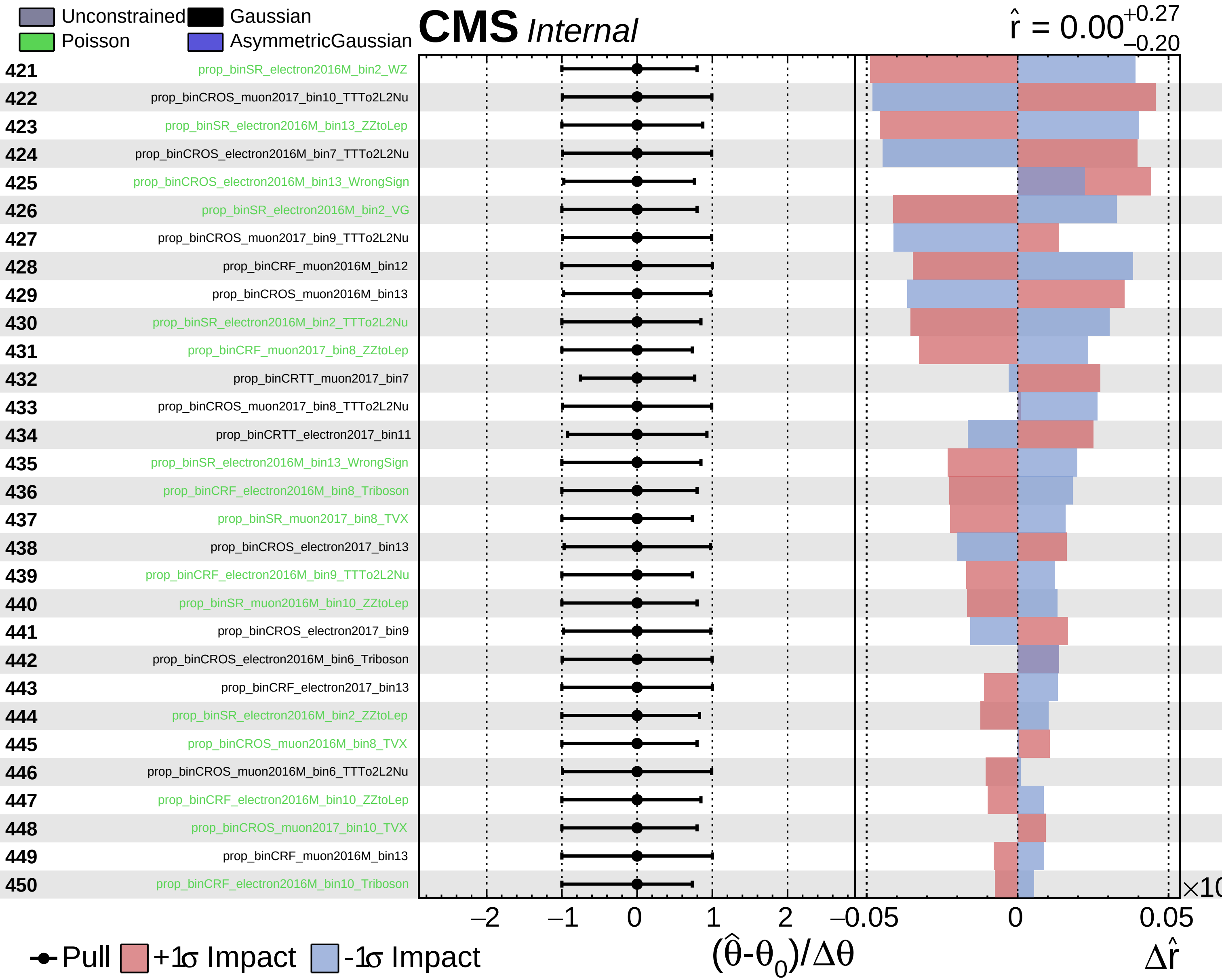

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
331 prop_binCROS_electron2016M_bin11
332 mischarge_electron_2017
333 prop_binCRF_muon2016M_bin7_WrongSign
334 prop_binCRTT_muon2017_bin12
335 prop_binCRTT_muon2016M_bin8
336 tau_vsjet_2018
337 prop_binCRTT_muon2018_bin6
338 prop_binCROS_muon2018_bin12
339 prop_binCRTT_muon2016M_bin9
340 prop_binCRTT_muon2016M_bin5
341 prop_binSR_electron2016M_bin13_Triboson
342 prop_binSR_electron2018_bin0
343 prop_binCROS_electron2016M_bin8
344 prop_binSR_muon2016M_bin10_Triboson
345 prop_binCRTT_electron2018_bin11
346 prop_binCRTT_electron2018_bin8
347 prop_binCRF_electron2016M_bin11
348 prop_binCRTT_muon2018_bin10
349 prop_binCROS_electron2017_bin8
350 prop_binCRTT_electron2017_bin9
351 prop_binCRTT_electron2016M_bin10
352 prop_binSR_electron2017_bin11_TVX
353 prop_binCRF_muon2016M_bin9_WrongSign
354 prop_binCRTT_electron2018_bin3
355 prop_binCRTT_muon2017_bin0
356 prop_binCROS_electron2016M_bin7_Fake
357 prop_binCROS_muon2018_bin11
358 prop_binCRF_electron2016M_bin12
359 prop_binCRTT_muon2018_bin2
360 prop_binCRF_electron2017_bin12
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



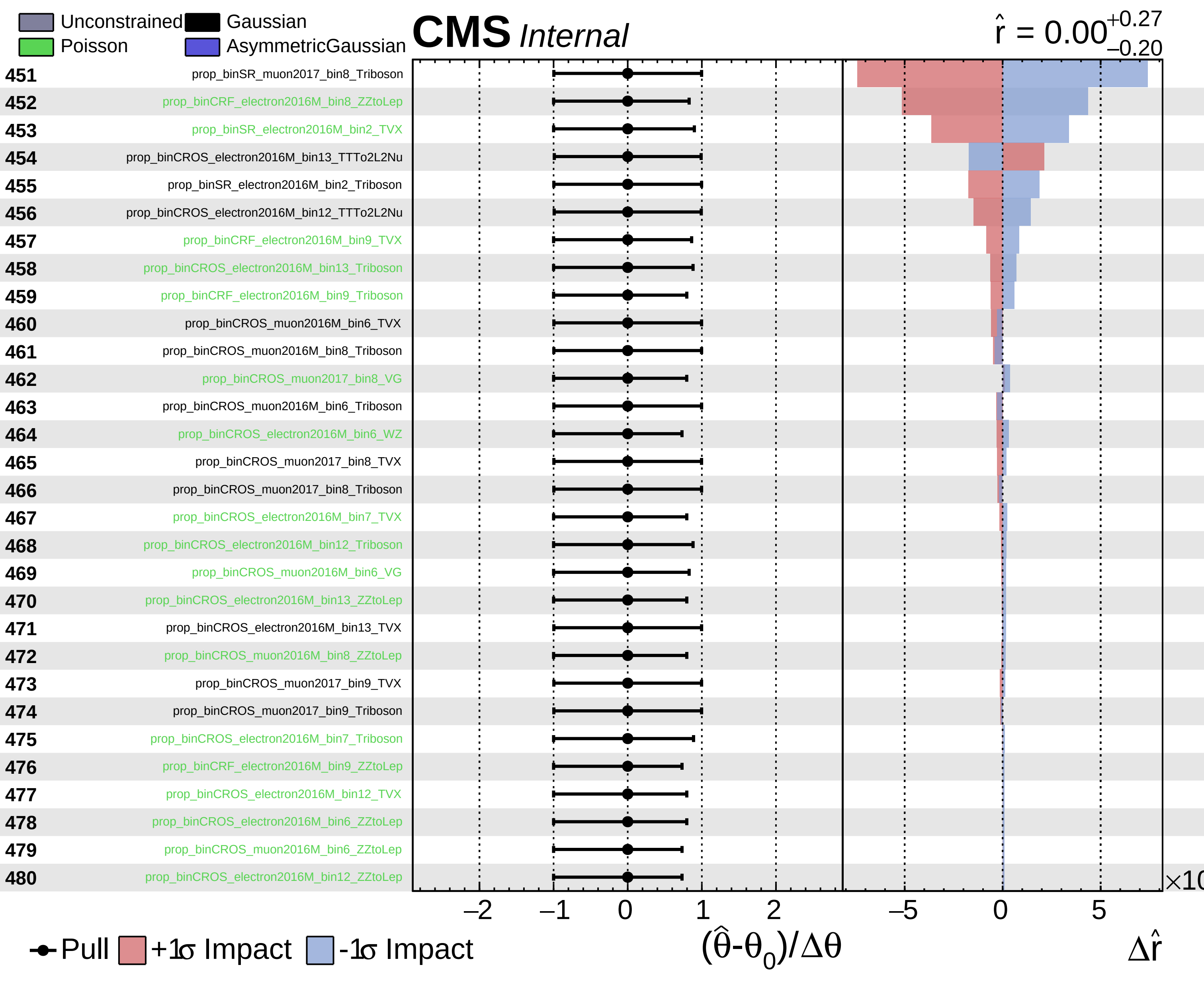
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
361 prop_binCRF_muon2017_bin0
362 prop_binCRTT_electron2016M_bin12
363 prop_binCROS_electron2018_bin4
364 prop_binSR_muon2016M_bin10_TVX
365 prop_binCRTT_electron2016M_bin7
366 prop_binCRF_electron2017_bin11
367 prop_binCROS_electron2017_bin7_Fake
368 prop_binCRTT_muon2018_bin13
369 prop_binCRF_muon2018_bin12
370 prop_binCRTT_muon2017_bin11
371 prop_binCROS_electron2018_bin9
372 prop_binCRTT_electron2018_bin9
373 prop_binSR_electron2017_bin11_WrongSign
374 prop_binCROS_muon2017_bin13
375 prop_binCROS_muon2016M_bin10
376 prop_binCRF_muon2017_bin12
377 prop_binCRTT_muon2017_bin1
378 prop_binCRTT_muon2017_bin4
379 prop_binCRF_electron2018_bin12
380 prop_binSR_muon2016M_bin0
381 prop_binCRTT_muon2017_bin9
382 prop_binCROS_electron2017_bin12
383 lep_2017
384 prop_binCRTT_electron2017_bin4
385 prop_binCRTT_electron2016M_bin5
386 pu
387 prop_binCRTT_muon2016M_bin10
388 prop_binCROS_electron2018_bin10
389 prop_binCRTT_electron2016M_bin11
390 prop_binCROS_muon2016M_bin6_Fake
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



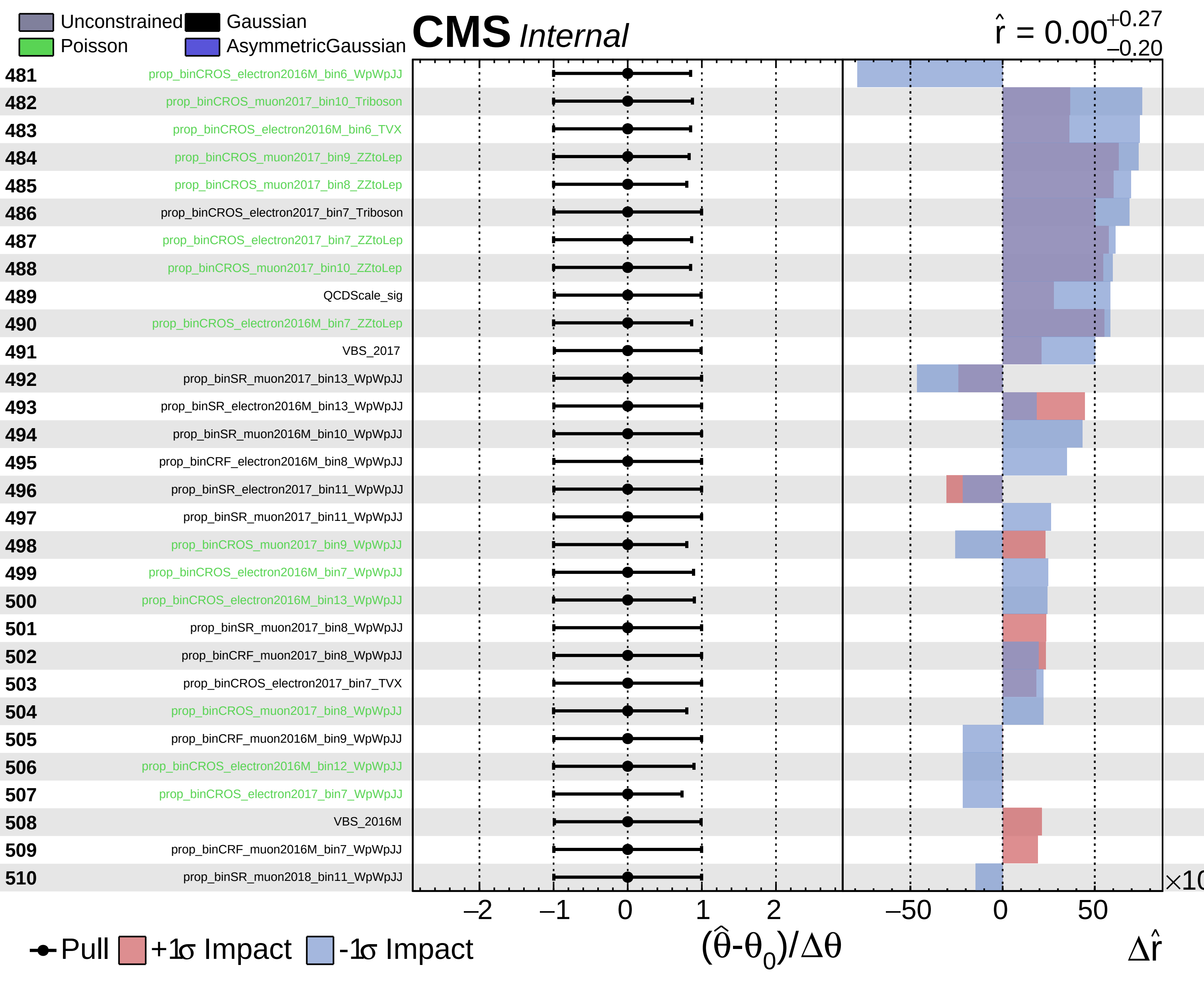
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
391 puID
392 prop_binCROS_muon2016M_bin11
393 prop_binCROS_muon2016M_bin8_TTTo2L2Nu
394 prop_binCROS_muon2017_bin9_Fake
395 prop_binCROS_muon2016M_bin12
396 prop_binCRTT_electron2016M_bin13
397 prop_binCRTT_muon2016M_bin13
398 prop_binCRF_muon2017_bin8_Triboson
399 prop_binCRTT_muon2016M_bin12
400 prop_binCRTT_muon2016M_bin11
401 prop_binSR_electron2017_bin11_Triboson
402 prop_binCROS_muon2017_bin12
403 prop_binCRTT_electron2017_bin1
404 prop_binCRF_muon2016M_bin7_Triboson
405 prop_binCRTT_electron2017_bin13
406 lep_2016M
407 prop_binCRF_muon2016M_bin9_TVX
408 prop_binCROS_electron2016M_bin6_TTTo2L2Nu
409 prop_binCRTT_muon2017_bin8
410 prop_binCROS_electron2017_bin7_TTTo2L2Nu
411 prop_binCROS_electron2016M_bin10
412 prop_binCRF_muon2016M_bin9_Triboson
413 prop_binCRF_electron2016M_bin8_TVX
414 prop_binCROS_electron2017_bin11
415 prop_binCRTT_electron2017_bin12
416 prop_binCRTT_electron2016M_bin8
417 prop_binCRF_electron2016M_bin10_TVX
418 prop_binSR_muon2017_bin8_TTTo2L2Nu
419 prop_binCROS_muon2018_bin13
420 prop_binCROS_muon2017_bin10_Fake
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

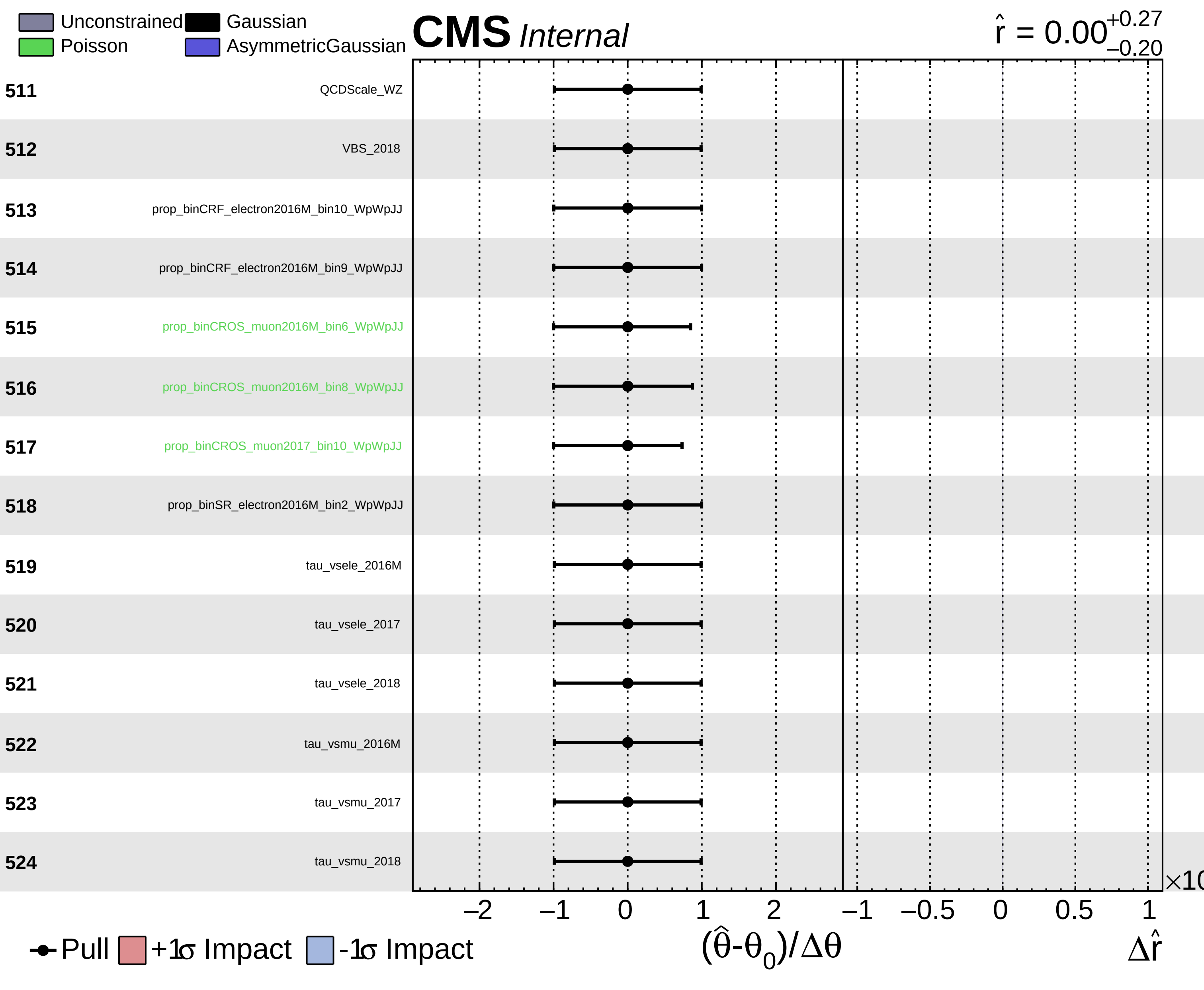


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
421 prop_binSR_electron2016M_bin2_WZ
422 prop_binCROS_muon2017_bin10_TTTo2L2Nu
423 prop_binSR_electron2016M_bin13_ZZtoLep
424 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
425 prop_binCROS_electron2016M_bin13_WrongSign
426 prop_binSR_electron2016M_bin2_VG
427 prop_binCROS_muon2017_bin9_TTTo2L2Nu
428 prop_binCRF_muon2016M_bin12
429 prop_binCROS_muon2016M_bin13
430 prop_binSR_electron2016M_bin2_TTTo2L2Nu
431 prop_binCRF_muon2017_bin8_ZZtoLep
432 prop_binCRTT_muon2017_bin7
433 prop_binCROS_muon2017_bin8_TTTo2L2Nu
434 prop_binCRTT_electron2017_bin11
435 prop_binSR_electron2016M_bin13_WrongSign
436 prop_binCRF_electron2016M_bin8_Triboson
437 prop_binSR_muon2017_bin8_TVX
438 prop_binCROS_electron2017_bin13
439 prop_binCRF_electron2016M_bin9_TTTo2L2Nu
440 prop_binSR_muon2016M_bin10_ZZtoLep
441 prop_binCROS_electron2017_bin9
442 prop_binCROS_electron2016M_bin6_Triboson
443 prop_binCRF_electron2017_bin13
444 prop_binSR_electron2016M_bin2_ZZtoLep
445 prop_binCROS_muon2016M_bin8_TVX
446 prop_binCROS_muon2016M_bin6_TTTo2L2Nu
447 prop_binCRF_electron2016M_bin10_ZZtoLep
448 prop_binCROS_muon2017_bin10_TVX
449 prop_binCRF_muon2016M_bin13
450 prop_binCRF_electron2016M_bin10_Triboson
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
451 prop_binSR_muon2017_bin8_Triboson
452 prop_binCRF_electron2016M_bin8_ZZtoLep
453 prop_binSR_electron2016M_bin2_TVX
454 prop_binCROS_electron2016M_bin13_TTTo2L2Nu
455 prop_binSR_electron2016M_bin2_Triboson
456 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
457 prop_binCRF_electron2016M_bin9_TVX
458 prop_binCROS_electron2016M_bin13_Triboson
459 prop_binCRF_electron2016M_bin9_Triboson
460 prop_binCROS_muon2016M_bin6_TVX
461 prop_binCROS_muon2016M_bin8_Triboson
462 prop_binCROS_muon2017_bin8_VG
463 prop_binCROS_muon2016M_bin6_Triboson
464 prop_binCROS_electron2016M_bin6_WZ
465 prop_binCROS_muon2017_bin8_TVX
466 prop_binCROS_muon2017_bin8_Triboson
467 prop_binCROS_electron2016M_bin7_TVX
468 prop_binCROS_electron2016M_bin12_Triboson
469 prop_binCROS_muon2016M_bin6_VG
470 prop_binCROS_electron2016M_bin13_ZZtoLep
471 prop_binCROS_electron2016M_bin13_TVX
472 prop_binCROS_muon2016M_bin8_ZZtoLep
473 prop_binCROS_muon2017_bin9_TVX
474 prop_binCROS_muon2017_bin9_Triboson
475 prop_binCROS_electron2016M_bin7_Triboson
476 prop_binCRF_electron2016M_bin9_ZZtoLep
477 prop_binCROS_electron2016M_bin12_TVX
478 prop_binCROS_electron2016M_bin6_ZZtoLep
479 prop_binCROS_muon2016M_bin6_ZZtoLep
480 prop_binCROS_electron2016M_bin12_ZZtoLep
−2 −1 0 1 2
−5 0 5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.27}_{-0.20}$
481 prop_binCROS_electron2016M_bin6_WpWpJJ
482 prop_binCROS_muon2017_bin10_Triboson
483 prop_binCROS_electron2016M_bin6_TVX
484 prop_binCROS_muon2017_bin9_ZZtoLep
485 prop_binCROS_muon2017_bin8_ZZtoLep
486 prop_binCROS_electron2017_bin7_Triboson
487 prop_binCROS_electron2017_bin7_ZZtoLep
488 prop_binCROS_muon2017_bin10_ZZtoLep
489 QCDScale_sig
490 prop_binCROS_electron2016M_bin7_ZZtoLep
491 VBS_2017
492 prop_binSR_muon2017_bin13_WpWpJJ
493 prop_binSR_electron2016M_bin13_WpWpJJ
494 prop_binSR_muon2016M_bin10_WpWpJJ
495 prop_binCRF_electron2016M_bin8_WpWpJJ
496 prop_binSR_electron2017_bin11_WpWpJJ
497 prop_binSR_muon2017_bin11_WpWpJJ
498 prop_binCROS_muon2017_bin9_WpWpJJ
499 prop_binCROS_electron2016M_bin7_WpWpJJ
500 prop_binCROS_electron2016M_bin13_WpWpJJ
501 prop_binSR_muon2017_bin8_WpWpJJ
502 prop_binCRF_muon2017_bin8_WpWpJJ
503 prop_binCROS_electron2017_bin7_TVX
504 prop_binCROS_muon2017_bin8_WpWpJJ
505 prop_binCRF_muon2016M_bin9_WpWpJJ
506 prop_binCROS_electron2016M_bin12_WpWpJJ
507 prop_binCROS_electron2017_bin7_WpWpJJ
508 VBS_2016M
509 prop_binCRF_muon2016M_bin7_WpWpJJ
510 prop_binSR_muon2018_bin11_WpWpJJ
−2
−1
0
1
2
−50
0
50
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.27 −0.20
511 QCDScale_WZ
512 VBS_2018
513 prop_binCRF_electron2016M_bin10_WpWpJJ
514 prop_binCRF_electron2016M_bin9_WpWpJJ
515 prop_binCROS_muon2016M_bin6_WpWpJJ
516 prop_binCROS_muon2016M_bin8_WpWpJJ
517 prop_binCROS_muon2017_bin10_WpWpJJ
518 prop_binSR_electron2016M_bin2_WpWpJJ
519 tau_vsele_2016M
520 tau_vsele_2017
521 tau_vsele_2018
522 tau_vsmu_2016M
523 tau_vsmu_2017
524 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂